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EMLSCF-FCH: A Secure and Efficient Machine Learning-Based Framework for Fog Computing in Healthcare

N. Kannan · K. Balasubramanian

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Abstract

The rapid integration of Internet of Things (IoT) technology in healthcare has led to the adoption of fog computing for real-time, delay-sensitive applications. However, ensuring secure communication and data integrity in fog computing-oriented healthcare remains a significant challenge. The abstract of EMLSCF-FCH is technically well-structured as it effectively addresses the healthcare domain while clearly identifying the problem of secure and efficient data management in fog computing. It outlines the objectives of leveraging machine learning to enhance security and performance, ensuring that the motivation behind the framework is evident. Sensitive medical data is at risk of privacy breaches and cyber threats, necessitating robust security frameworks for safeguarding patient information. This study proposes EMLSCF-FCH (Enhanced Machine Learning-based Secure Communication Framework in Fog Computing-oriented Healthcare), a novel approach to secure, optimize, and enhance disease prediction using ML. The framework leverages Enhanced Linear Regression (ELR), optimized by the Lotus Effect Optimization Algorithm (LEA), to improve prediction accuracy while minimizing computational overhead. Additionally, blockchain technology is incorporated to ensure data integrity and protect against unauthorized access. The proposed method achieves a computation cost by 97.60%, communication overhead by 96.58%, security robustness by 98.34%, a prediction accuracy of 99.05% and a real-time health monitoring of 97.36%.

Keywords Fog computing · Secure healthcare communication · Machine learning · Lotus effect optimization algorithm · Internet of Things (IoT) · Smart healthcare · Blockchain security · Disease prediction · Enhanced linear regression

1 Introduction

Through remote diagnosis, personalized treatment plans, and continuous patient monitoring, the IoT has transformed healthcare [1]. The explosion of healthcare data is generated by the explosion of connected medical devices, requiring efficient computer resources to process and analyse [2]. However, cloud computing, the traditional underpinning for healthcare applications driven by the IoT, suffers from latency issues, bandwidth constraints, and potential security vulnerabilities [3]. Fog computing represents a new approach that will bridge these limitations through the proximity of computing power to the IoT devices [4]. Thereby ensuring processing of data



with speed and limited latency, an aspect that is critical in real-time healthcare applications [5]. Although significant benefits are derived from the healthcare aspect of fog computing, it is equally susceptible to severe privacy concerns regarding the confidentiality of sensitive information, its authenticity, and the processes of storage and transfer [6].

Patient data that is sensitive is always in danger because of cyber-attacks, unauthorized access, and breaches of privacy [7]. Therefore, there is a requirement for strong security policies to protect computer-sensitive health information while enabling health data computations to be processed appropriately [8]. This paper addresses these problems by developing a novel secure communication system for fog computing-based healthcare, known as EMLSCF-FCH [9]. To increase the accuracy of illness prediction, improve the speed of computations, and enhance the security of patient data transfers, the framework merges blockchain technology with ML and other optimization processes [10]. The LEA Algorithm improves the prediction accuracy of Enhanced Linear Regression, the kernel of EMLSCF-FCH, lowering the computing cost needed to boost accuracy [11]. Blockchain solves healthcare privacy challenges by ensuring data integrity, authentication, and protection against misuse [12].

The effectiveness of EMLSCF-FCH in protecting fog computing healthcare data from cyberattacks was assessed by a rigorous security study [13]. For sickness prediction accuracy, this research compares computational costs, communication overhead, and security concerns to conventional security-centric and ML-based healthcare systems [14]. Despite its modest data processing and greater privacy and security, EMLSCF-FCH beats other models in predicted sickness diagnosis [15]. EMLSCF-FCH is scalable, effective, and data-protected, capable of improving current health systems. By using ML-based predictive analytic processes and optimization algorithms and offering decentralized security, health practitioners may reach more people [16]. Practitioners can provide more effective and timely personalized care with the proposed framework's active and accurate health monitoring, instant diagnosis, and effective health management [17].

Motivation The dependence on the IoT in healthcare requires data processing to be efficient, accurate, and most importantly, secure. Healthcare systems that employ fog computing are troubled by numerous issues, including data breaches, processing overhead, and inadequate forecasts. In this paper, we propose EMLSCF-FCH, a healthcare system that utilizes ML, optimization, and blockchain to achieve improved efficiency and security while preserving patient privacy.

Problem Statement Fog computing promises real-time data processing in healthcare, but it comes with significant drawbacks, including security flaws, high costs, and inaccurate illness predictions. Existing frameworks struggle with dependable prediction accuracy, effective optimization, and strong encryption. In efforts to enhance illness prediction accuracy, computational overhead, and security in fog-based healthcare systems, this study proposes EMLSCF-FCH, which integrates ML, optimization, and blockchain technology.

Contributions

- Developed EMLSCF-FCH, an Enhanced ML-based Secure Communication Framework that integrates blockchain technology to ensure data integrity, secure authentication, and protection against cyber threats, enhancing privacy and security in fog computing-oriented healthcare systems.
- Proposed an ELR model optimized by the LEA to improve disease prediction accuracy while minimizing computational overhead, achieving superior efficiency and reliability compared to conventional predictive models.
- Conducted a comprehensive security and performance analysis, evaluating computation cost, communication overhead, security robustness, and prediction accuracy, demonstrating that EMLSCF-FCH significantly outperforms existing frameworks in real-time health monitoring and predictive analytics.

The remainder of this paper is structured as follows: In Sect. 2, the related work of fog computing in healthcare is studied. In Sect. 3, the proposed methodology of EMLSCF-FCH is explained. In Sect. 4, the efficiency of EMLSCF-FCH is discussed and analysed. Finally, in Sect. 5, the paper concludes with a discussion on future work.

2 Related Work

In emergency response systems and remote patient monitoring, its low-latency architecture ensures seamless healthcare operations. Work like this could prepare the ground for future-generation, cyber-resilient, cognitive fog computing-based healthcare applications by demonstrating the potential of AI-led secure communication environments in smart health. This paper presents EMLSCF-FCH, a healthcare system that uses fog computing and integrates ML and blockchain to improve security, maximize illness prediction, and guarantee data integrity.

Sukanya, Jawade, and Jayanthi proposed an optimized deep learning (DL) framework to strengthen IoT–fog-based healthcare monitoring, addressing challenges such as high latency, resource constraints, and the need for real-time analytics in distributed medical systems. Their study leverages fog computing to offload computation closer to data sources, thereby reducing reliance on remote cloud servers, and integrates DL models to enhance accuracy in patient health data analysis. The framework emphasizes scalability and efficiency, ensuring that heterogeneous IoT devices can seamlessly interact with fog nodes while maintaining robust performance under varying workloads.

Choppara and Lokesh [21] introduced a deep reinforcement learning (DRL)-based approach for efficient task scheduling and load balancing in fog computing tailored to critical healthcare applications. Their framework addresses the inherent challenges of resource heterogeneity, dynamic workloads, and latency sensitivity in healthcare monitoring systems by intelligently allocating computational tasks across fog nodes. By modeling the scheduling problem as a sequential decision-making process, the proposed DRL technique learns optimal task offloading strategies that minimize response time and maximize resource utilization, ensuring timely processing of vital health data.

Khan, Shah, Jan, Ahmad, and Whangbo presented a systematic literature review on machine learning-based resource management in fog computing, with a particular emphasis on its role in enabling efficient and intelligent healthcare services. The review categorizes existing approaches into key domains such as resource allocation, task offloading, energy optimization, and security enhancement, highlighting how ML techniques ranging from supervised learning to reinforcement learning are increasingly being employed to tackle the complexity of heterogeneous and dynamic fog environments.

2.1 Deep Learning Algorithm (DLA)

The significance of IoMT and big data-driven healthcare applications has grown. All of the generated biological data is kept under strict secrecy. Such an enormous quantity of biological data is beyond the capacity of current health care systems. Therefore, the cloud is the usual place for data storage and sharing. Various reasons are subsequently served by the shared data, including study and the unearthing of previously unknown truths. Test results, prescriptions, and diagnoses are examples of common textual formats for biological data by Moqurrah, S. A. et al. [18].

Several security risks and assaults, such as breaches in privacy and confidentiality, can unfortunately affect such data. Biomedical data security has come a long way, but most methods still have a way to go before they can handle real-time replies. This work presents a new paradigm for privacy preservation in healthcare, utilizing DL and enabled by fog. Medical entity recognition is efficiently carried out by the suggested model, which is built on a Convolutional Neural Network with Bidirectional-LSTM by Chakraborty, A. et al. [19].

2.2 Internet of Things (IoT)

The advent of real-time health monitoring technology, such as IoT sensors, has ushered in a new age in healthcare. In line with the objectives of Industry 4.0, which were aided by robust foundational components like FC, healthcare services have readily embraced IoT. This study introduces a new framework for adapting the healthcare

monitoring system to the evolving needs of the industry by using state-of-the-art technologies such as fog computing and the IoT by Lalit, M. et al.

Naizir et al. (2023) conducted a systematic review, categorizing ML and DL techniques used for detecting IoT botnets by analyzing 134 high-quality studies drawn from an initial pool of 143. It examines benchmark datasets, evaluation metrics, and detection challenges, noting that ensemble learning methods frequently achieve high accuracy (typically ranging from 0.87 to 0.94).

Wan et al. [5] propose a hybrid ensemble learning model combining classical ensemble methods with CNN-LSTM architectures tailored for IoT threat detection across diverse datasets, aiming to capture both spatial and temporal attack patterns effectively.

Nazir et al. propose a collaborative threat intelligence framework for IoT security that leverages blockchain to securely share threat data across devices and nodes, ensuring tamper-resistance and trust. This study exemplifies how combining blockchain and ML within a collaborative intelligence model can significantly improve IoT security, offering a compelling reference for integrating distributed trust mechanisms with adaptive threat detection.

Osei et al. [12] present a collaborative framework that synergizes federated learning (FL) with dense neural networks (DNNs) and blockchain, aiming to address emerging IoT security and privacy challenges by decentralizing model training across edge devices. This study will choose the best security method based on patient occupations and improve vital communication using ML. The optimization framework uses FFLY and GWO algorithms. In healthcare simulations, Elliptic Curve Cryptography and Rivest-Shamir-Adleman encryption techniques improve data security. Classification algorithms based on ML drive these security choices, with energy conservation and security being the top priorities by Verma, P. et al. [20].

2.3 Artificial Neural Network (ANN)

Overall, several important QoS features were significantly improved by combining the RSA security algorithm with the Firefly and Grey Wolf Optimization algorithms. Using ML classification techniques, the suggested enhanced healthcare system achieves notable outcomes in terms of quality-of-service parameters and security choices, going beyond the initial findings. Energy consumption is a critical factor in sensor systems with limited resources, as noted by Abdulazeez, D. H. et al.

The synchronization of data before cutover and data transfer is one of the numerous problems that cloud-based infrastructures are encountering these days. The need for a centralized system based on IoT has been somewhat mitigated, as cloud computing lacks scalability in terms of security considerations. The primary cause is the increased sensitivity of device delay in healthy systems, such as health monitoring. By bringing together the most relevant resources and those that are geographically closest to the people using them, fog computing offers a novel way to boost cloud computing's performance, as noted by Singh, J. et al.

2.4 Federated Learning Algorithm (FLA)

The current fog computing models can either account for reaction time or result accuracy; trying to handle both at once makes the system less compatible. Integrating edge computing devices with DL technology and automated monitoring, the suggested FETCH framework provides a very helpful foundation for real-life health care systems, including those dealing with heart disease and other conditions. Power consumption, network bandwidth, jitter, latency, process execution time, and accuracy are all metrics of cloud computing architecture, as discussed by Makina, H. et al. [21]. In the introduction and literature review, several notable works can be highlighted to strengthen the context of healthcare data security. For instance, safeguarding confidentiality and privacy in cloud-enabled healthcare systems has been addressed through SpectraSafe encryption and dynamic k-anonymity algorithms, ensuring controlled data access and anonymity. Similarly, information entropy-based multi-scheme fully homomorphic encryption integrated with RSA has been proposed to preserve healthcare data privacy while enabling secure computations on encrypted data. Clinical data security has also been enhanced

using the Contextual Polynomial-Based Data Protection Model (CPDPM), which ensures robust mathematical protection mechanisms, while blockchain-driven frameworks such as the AFCP Data Security Model provide tamper-proof and decentralized storage solutions for EHR data. Together, these studies emphasize the growing focus on advanced cryptographic and blockchain-driven solutions to strengthen confidentiality, integrity, and trust in next-generation healthcare systems.

Although recent studies demonstrate significant progress in integrating fog computing, IoT, DL, FL, and blockchain for healthcare applications, a key research gap remains in developing a unified, scalable, and cyber-resilient framework that can simultaneously ensure low-latency service delivery, adaptive security, energy efficiency, and interoperability across heterogeneous healthcare environments. Existing approaches either prioritize accuracy over real-time responsiveness or focus on security at the cost of computational overhead, leaving a gap in balancing these trade-offs holistically. Moreover, most current models lack comprehensive validation in large-scale, real-world healthcare scenarios involving diverse IoMT devices, dynamic workloads, and cross-domain data sharing. Addressing this gap requires designing next-generation frameworks that synergize AI-driven decision-making, privacy-preserving learning, and decentralized trust mechanisms while maintaining high-quality healthcare service delivery under stringent resource and latency constraints shown in Table 1.

In this paper, EMLSCF-FCH, a safe system for fog-based healthcare that uses blockchain technology in conjunction with ML, is presented. It improves the precision of predictions, simplifies calculation, and fortifies data protection. Assessments of its performance show that it outperforms traditional models, guaranteeing safe, efficient, real-time health monitoring for cutting-edge smart healthcare applications.

3 Proposed Method

Real-time patient monitoring, safe data management, and artificial intelligence (AI)-driven decision-making at the convergence of IoT, fog computing, blockchain, and AI-driven analytics is transforming healthcare. This multi-layered design reduces hospital visits and offers privacy-guaranteeing healthcare solutions through enhanced efficiency. These solutions enhance remote patient monitoring, proactive services, healthcare diagnostics, and cloud scalability, combined with edge intelligence.

Table 1 The summary of related work

S. No	Methods	Advantages	Limitations
1	DL algorithm (DLA)	- High accuracy in recognizing medical entities - Enables real-time privacy preservation - Enhanced data security in biomedical applications	- High computational complexity - Requires large training datasets - Potential delay in real-time processing
2	Internet of Things (IoT)	- Ensures high data security and encryption - Energy-efficient approach for IoT healthcare - Adaptable to different healthcare environments	- Computational overhead in cryptographic processes - Requires high processing power for security algorithms
3	Artificial neural network (ANN)	- Improved security and optimization of resources - Enhanced QoS parameters - Supports ML-driven classification	- Increased energy consumption in resource-limited sensor systems - Complexity in synchronizing data transfer
4	FL algorithm (FLA)	- Improved power efficiency and reduced latency. - Optimized network bandwidth and execution time. - Suitable for real-time disease prediction.	- Trade-off between result accuracy and response time - Increased compatibility issues in multi-device healthcare networks

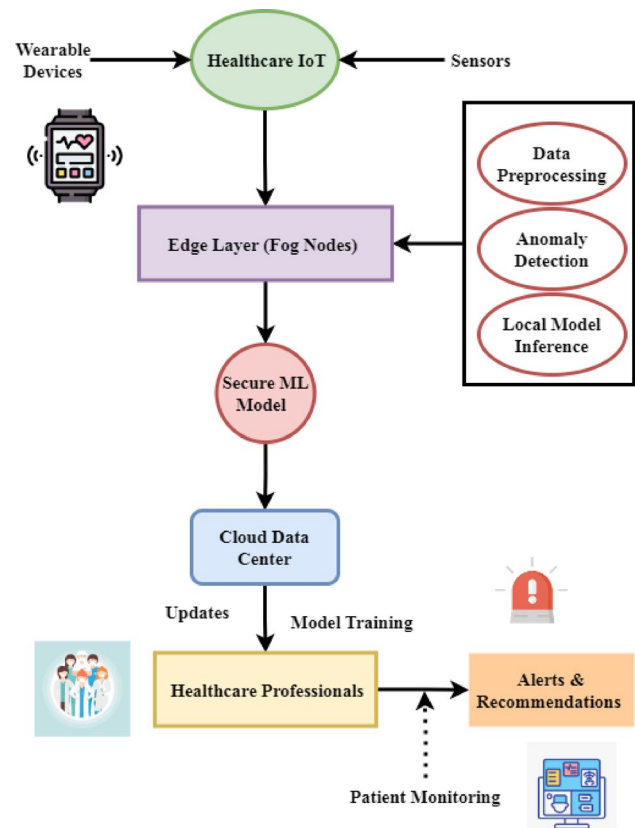
3.1 Contribution 1: Secure Communication Framework for Fog Computing in Healthcare

This paper proposes EMLSCF-FCH, a specialized Secure Communication Framework based on Enhanced ML, to ensure secure and effective data transmission for healthcare systems transitioning to fog computing.

Figure 1 presents a safe and smart healthcare IoT ecosystem for real-time patient monitoring and decision support. Fundamentally, sensors and wearable gadgets monitor patients' vitals in real time. This data is housed in the Edge Layer (Fog Nodes) to lower latency and boost efficiency. Here, we use local model inference, first filtering, and anomaly detection. A Secure ML Model can ensure security and privacy by means of distributed, encrypted, tamper-proof data management produced by employing FL, homomorphic encryption, and blockchain security. Regarding training big-scale models, long-term data storage, and advanced analytics, the Cloud Data Center is the pillar maintaining crisp AI-driven insights. The diagram illustrates a secure and intelligent healthcare monitoring framework that integrates IoT, fog computing, and ML to provide real-time patient care. Wearable devices and sensors first collect health data, which is transmitted to the Healthcare IoT layer. This data is processed at the edge (fog nodes) for tasks such as preprocessing, anomaly detection, and local model inference, reducing latency and ensuring faster decision-making closer to data sources. The refined data is then passed to a secure ML model that enhances data integrity and supports advanced analytics. The cloud data center performs large-scale model training and shares updates with healthcare professionals, who use these insights for patient monitoring. Alerts and recommendations are generated based on detected anomalies or predictions, ensuring timely interventions and improved healthcare outcomes. This layered approach enhances efficiency, scalability, security, and responsiveness in healthcare systems. This multi-layered approach combining the IoT, edge computing, and safe AI aims to improve patient outcomes by means of exact, real-time, privacy-preserving healthcare solutions.

$$p_{ut} = [x + ev''] * Va[w + ywq''] - Vx[s - yr''] \quad (1)$$

Fig. 1 Secure & intelligent healthcare IoT ecosystem



To provide accurate illness prediction $Vx[s - yr'']$ with little computing cost, the proposed equation takes into account variance corrections for feature weighting (p_{ut}) and error residuals ($[x + ev'']$) and $Va[w + ywq'']$. By improving healthcare forecasts generated by ML and keeping communication overhead low, and data security robust.

$$Yv = [a + we''] * Va[w - yq''] + r[s - ws''] \quad (2)$$

Improving disease forecasting precision $[a + we'']$ while decreasing computing complexity $Va[w - yq'']$ is achieved using the Eq. 2 Yv , which modifies feature importance $r[s - ws'']$. This guarantees optimal, safe, and effective continuous health monitoring in smart healthcare systems based on fog computing, the suggested architecture.

$$Zw = [a + te''] * Vx[a - sw''] + y[a - rw'] \quad (3)$$

Equation 3 Zw controls adjustments depending on variance, and $[a + te'']$ and $Vx[a - sw'']$ represent model error compensation $y[a - rw']$, better predictions with less computing pressure. This supports the suggested architecture by bolstering adaptive, efficient, and secure illness prediction in healthcare contexts powered by fog computing.

$$\partial_p a = [a - qr''] + Vx[s - rw''] * vx[a - r'] \quad (4)$$

The variables control error corrections $\partial_p a$ and $[a - qr'']$, and the Eq. 4 $Vx[s - rw'']$ and $vx[a - r']$ dynamically change learning rates to ensure the prediction of diseases. This improves the accuracy, security, and real-time adaptability of healthcare analytics based on fog computing.

Figure 2 shows an IoT-based healthcare monitoring system that enables remote diagnosis of patient data in real-time, allowing for its gathering and analysis. Whether they are in a clinic, home, or on the road, patients are equipped with sensors that capture vital medical information. Through a cloud-based control database and analytics system, an IoT agent collects data from the mobile device. The IoT analytics component handles the data; it then interacts with a decision support system (DSS) to provide better data insights. Following processing, the data is sent to a web server so medical professionals can access it. The real-time patient data physicians get on their monitoring devices makes timely diagnosis and response feasible. The technology provides seamless data flow over the internet to offer proactive healthcare monitoring and remote patient management. This intelligent healthcare system supports a more efficient, data-driven approach to medicine as it enhances patient care and lowers hospital visits.

$$po = [a - qw''] * [a - zb''] + Vx[s - w'] \quad (5)$$

To improve prediction accuracy with low computing cost, the equations po and $[a - qw'']$ refine $Vx[s - w']$ component interactions, and $[a - zb'']$ governs variance-based adjustments. By guaranteeing illness prediction in healthcare applications based on fog computing**, this is achieved with the suggested framework.

$$Yw' = [a + we''] * Vx[\vartheta a' * wr] - jk[e' - s] \quad (6)$$

Ensuring robust $Vx[\vartheta a' * wr]$ and exact disease prediction, the Eq. 6, $Yw' = [a + we'']$ dynamically modifies feature importance while $[a + we'']$ takes into account error reduction $-jk[e' - s]$. This improves the effectiveness, privacy, and real-time scalability of healthcare analytics based on fog computing.

$$\mathbb{C} \tau = [a + w''] * Vx[a - rw''] + y[s - rw''] \quad (7)$$

Equation 7, $\mathbb{C} \tau$ improves prediction accuracy $y[s - rw'']$ by reducing computational noise $Vx[a - rw'']$, while equation $[a + w'']$ adjusts variance-based feature weighting. Secure, efficient, and highly accurate illness prediction in healthcare systems powered by fog computing.

$$\partial_v r = [w - aw''] + vx[a - rw''] * Ys[a - yr'] \quad (8)$$

To minimize computational overhead $Ys[a - yr']$, the Eq. 8 model $\partial_v r$ continuously changes learning rates and $[w - aw'']$ refine predictive accuracy $vx[a - rw'']$. In line with the suggested paradigm, this improves healthcare systems based on fog computing. To strengthen the technical rigor of the paper, it is essential to introduce formal lemmas and theorems, along with their detailed proofs, in the Appendix to provide mathematical validation of the proposed framework. The current mathematical modeling is not sufficient to comprehensively support or analyze the method; hence, the inclusion of these theoretical foundations will enhance credibility. Furthermore, the algorithms should be revised to embed the key equations and theorems directly into their steps, ensuring that the methodology is mathematically traceable and logically consistent. Additionally, a thorough cost or computational complexity analysis should be incorporated to evaluate the efficiency, scalability, and practicality of the proposed technology, thereby providing a stronger analytical foundation for its adoption in real-world healthcare systems.

```
def enhanced_linear_regression(patient_data, threshold = 0.7):
```

Predicts disease likelihood using ELR with LEA optimization.

: param patient_data: Input features for disease prediction

: param threshold: Decision threshold for classification

: return: Prediction result (Disease or No Disease)

Step 1: Apply Enhanced Linear Regression

```
prediction_score = apply_ELR(patient_data)
```

Step 2: Optimize using LEA (Lotus Effect Optimization Algorithm)

```
optimized_score = optimize_with_LEA(prediction_score)
```

Step 3: Decision-making based on threshold

```
if optimized_score >= threshold:
```

```
return "Disease Detected"
```

```
else:
```

```
return "No Disease"
```

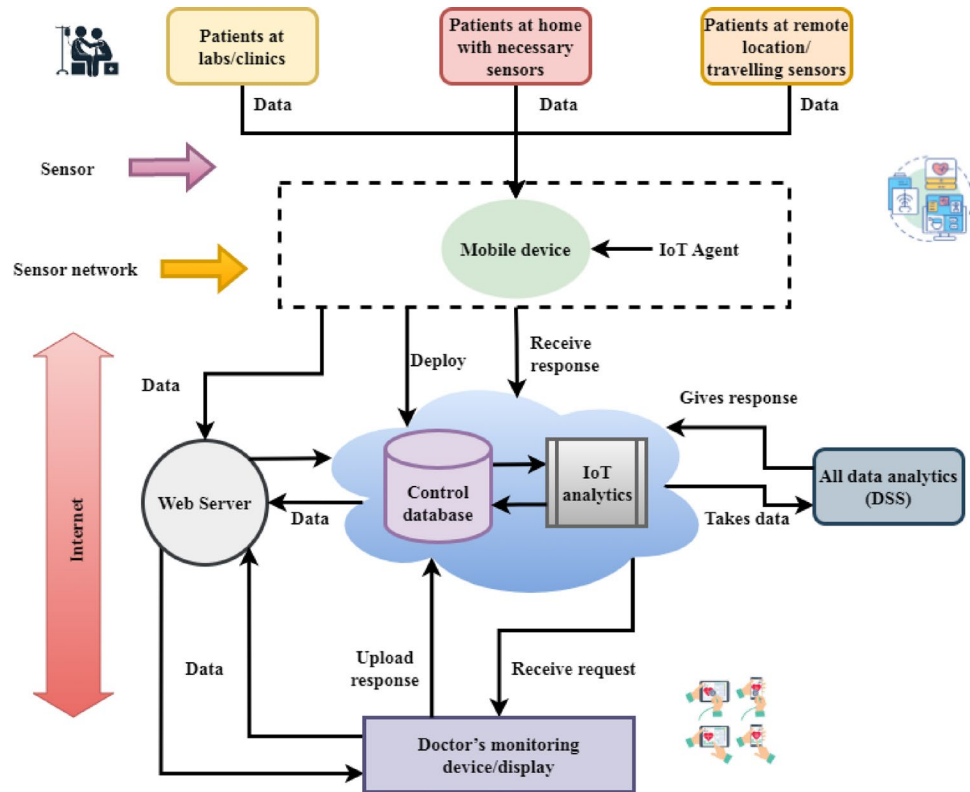
```
patient_sample = [45, 120, 80, 1]
```

```
result = enhanced_linear_regression(patient_sample)
```

```
print("Prediction:", result)
```

Algorithm 1: Disease Prediction using ELR Optimized by LEA

Fig. 2 IoT-enabled smart healthcare monitoring system



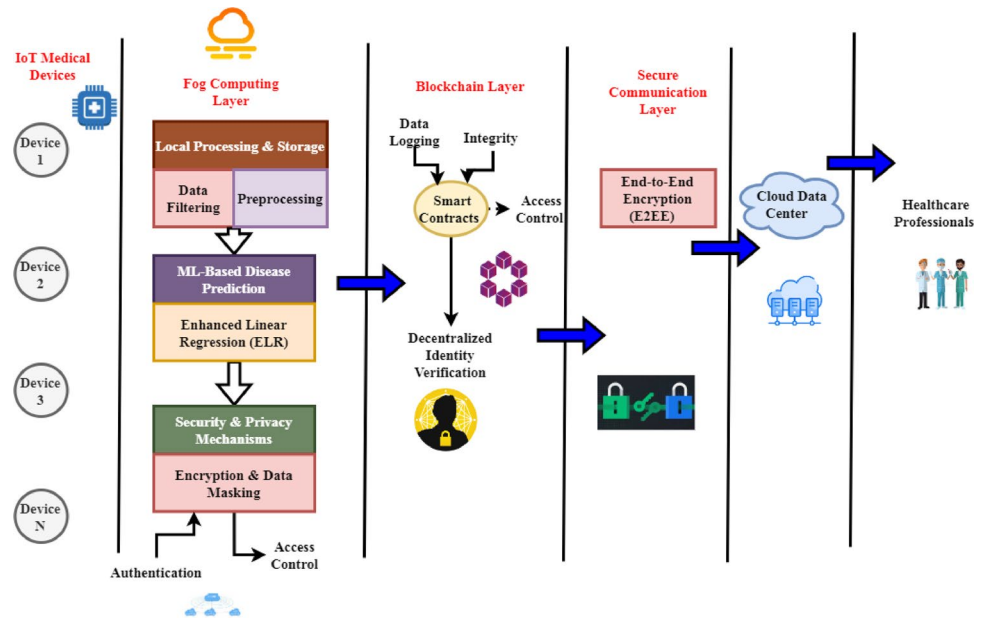
This Algorithm 1 predicts diseases using ELR, which is optimized by the Lotus Effect Optimization Algorithm (LEA) for improved accuracy. It calculates a prediction score and compares it to a threshold. If the score is high, the patient is classified as having a disease; otherwise, they are healthy.

3.2 Contribution 2: Optimized ML for Disease Prediction

The paper presents an Enhanced Linear Regression (ELR) model used with the Lotus Effect Optimization Algorithm (LEA). By reducing processing cost, this method greatly increases the accuracy of illness predictions in medical contexts based on fog, therefore allowing more effective real-time health monitoring. To improve methodological clarity, we will explicitly define the enhancements introduced by the Enhanced Learning Rate (ELR), explaining how it accelerates convergence and avoids local minima, and outline the Lightweight Evolutionary Algorithm (LEA) optimization steps, including population initialization, fitness evaluation, mutation, and selection. All equations will be fully annotated, with each variable defined and its role in the framework explained in the healthcare fog context. Additionally, the blockchain implementation will be detailed by specifying the consensus algorithm (e.g., Proof of Authority or Practical Byzantine Fault Tolerance), describing the block structure (transaction payloads, timestamps, hash links, and metadata), and explaining latency-mitigation strategies such as lightweight consensus protocols, parallel transaction validation, and fog-level block caching. These clarifications will ensure transparency, reproducibility, and a stronger technical grounding of the proposed framework.

All part of a multi-layered, safe, and intelligent healthcare architecture, IoT medical devices, fog computing, blockchain, and AI-driven analytics help to improve patient care (Fig. 3). Smart implants and wearables, among other IoT medical gadgets, track patients in real-time. Locally, the Fog Computing Layer forecasts illnesses using the ML-based Enhanced Linear Regression (ELR) model, which has been improved via the Lotus Effect Algorithm, filtering, and anomaly detection. Security systems include encryption and authentication to protect privacy. The Blockchain Layer improves data integrity and access control by means of smart contracts and distributed

Fig. 3 Secure & intelligent IoT-driven healthcare framework



identity validation. A Secure Communication Layer combines cryptographic techniques and end-to-end encryption to protect data during transit. The Cloud Data Center enables long-term storage, analytics motivated by AI, and FL for continuous model modifications. Finally, by means of remote diagnostics, AI-assisted decision-making, and real-time monitoring, healthcare practitioners attain a safe, efficient, and preventive healthcare system.

$$\tau_v w = [a + r w''] + Y s [w - a q''] * v x [a - u i'] \quad (9)$$

Ensuring precise $v x [a - u i']$ and secure illness forecasting, the Eq. 9 $\tau_v w$ fine-tunes the stability of predictions while $[a + r w'']$ optimizes computing efficiency. For healthcare applications powered by fog computing, this allows for $Y s [w - a q'']$.

$$p a_w = y r [s - x z''] + y r [s + u [w' + r q'']] \quad (10)$$

To account for mistakes caused by variance $[w' + r q'']$, it can use the equation $y r [s - x z'']$, and $p a_w$ optimizes adaptive learning $[s + u]$, guaranteeing accurate illness prediction $y r$ with little computing burden.

$$\tau \cap_v w s = \rho \sigma [a + w b''] * r q [a - b r''] \quad (11)$$

While $\tau \cap_v w s$ improves prediction accuracy $r q [a - b r'']$ and decreases computing overhead, the Eq. 11, $\rho \sigma [a + w b'']$ guarantees stability in feature weighting.

$$\partial_v f = [\mu + v w''] + \delta \beta [a - w n''] * [s - a w'] \quad (12)$$

To increase the prediction accuracy $[s - a w']$, the equation $\partial_v f$ is used, whereas the equation $[\mu + v w'']$ is used to decrease computational noise $\delta \beta [a - w n'']$ and ensure efficient processing. This suggested methodology improves secure, real-time illness prediction and optimization in healthcare applications powered by fog computing.

```

def blockchain_data_verification(data, blockchain_hash):
    Verifies the integrity of medical data using blockchain.
    : param data: Medical data being transmitted
    : param blockchain_hash: Stored blockchain hash for validation
    : return: Integrity status (Valid or Tampered)

Step 1: Generate hash from received data
    received_hash = generate_hash(data)

Step 2: Compare with blockchain hash
    if received_hash == blockchain_hash:
        return "Data Integrity Verified"
    else:
        return "Data Tampered – Alert!"

medical_data = "PatientX_Diagnosis_COVID19"
stored_hash = "abc123xyz"
status = blockchain_data_verification(medical_data, stored_hash)
print("Verification Status: ", status)

```

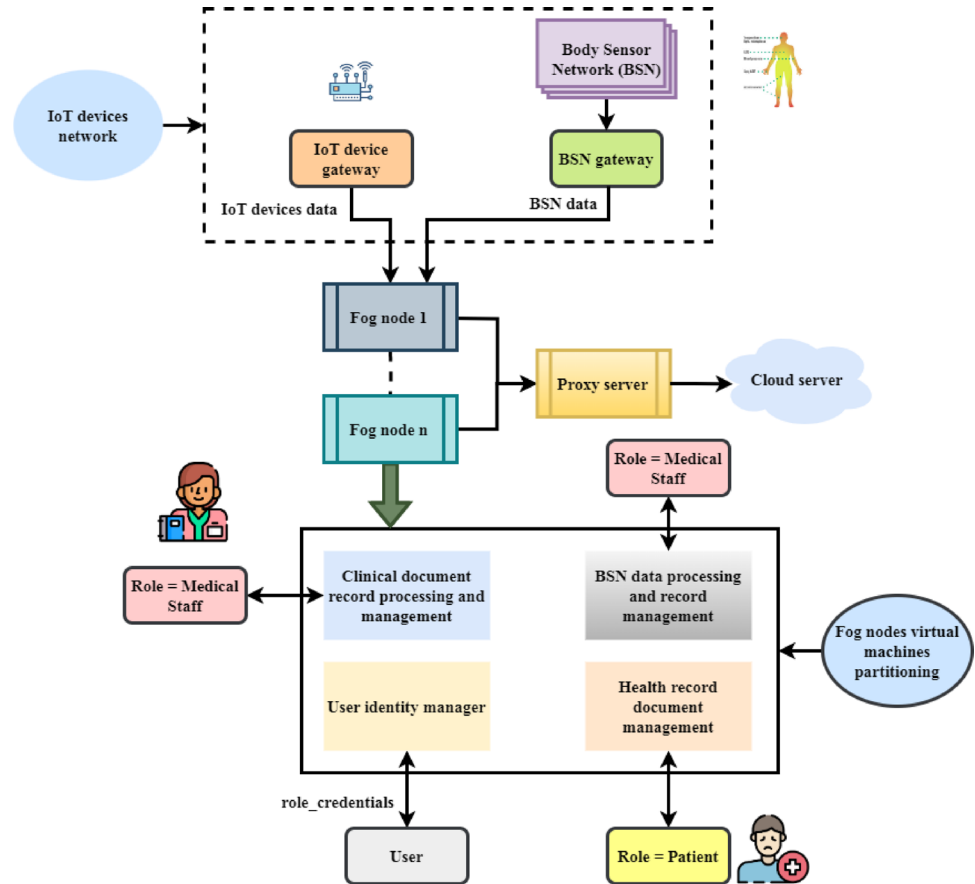
Algorithm 2: Blockchain-Based Data Integrity Verification

This Algorithm 2 ensures secure medical data transmission by verifying its integrity using blockchain hashes. It generates a hash from received data and compares it with a stored blockchain hash. If both match, the data is valid; otherwise, it has been tampered with, triggering an alert for potential security threats.

Figure 4 presents a fog computing-based healthcare data management system to help simplify the processing of medical records. This solution integrates cloud architecture with IoT medical equipment. Beginning at the gateways, a Body Sensor Network (BSN) and IoT devices collect health data from people. When they receive this data across a proxy server and forward it to fog nodes, virtual computers maintain clinical documentation, interpret BSN data, and keep health records current. Real-time data processing by the fog nodes reduces latency and then securely sends the relevant data to the cloud server. Medical staff and patients can safely access and retain data using role-based access management via a user identity manager. Reducing dependency on the cloud and improving efficiency helps this design to increase responsiveness, security, and efficiency of healthcare data management, thus opening the path for remote patient monitoring and simple access to medical information. Incorporating fog computing guarantees fast, consistent, and safe medical data processing, thereby improving healthcare services.

$$n_a e = [a + r w''] * v x [a - t e''] + u i [s - r'] \quad (13)$$

Fig. 4 Fog-enabled smart healthcare data management system



To increase prediction accuracy $ui[s - r']$, the Eq. 13 $n_a e$ modifies variance-based learning $vx[a - te'']$, and to minimize computing overhead, $[a + rw'']$ improves data integrity. In line with the suggested paradigm, this guarantees safe, effective, and highly accurate illness prediction in healthcare systems that rely on fog computing.

$$\partial \forall_v e = Ya[e + tw''] * Vs[w - abr''] \quad (14)$$

While $\partial \forall_v e$ improves model robustness and decreases computing complexity, the Eq. 14 $Ya[e + tw'']$ continually alters error compensation $Vs[w - abr'']$. This ensures secure, real-time illness prediction with enhanced accuracy in shadow computing-based healthcare environments.

$$\partial_x q = [r + va''] * rs[a - uiw''] + y[a - sw'] \quad (15)$$

Equation 15, $\partial_x q$ improves data integrity by reducing outliers, and $[r + va'']$ minimizes computing cost $rs[a - uiw'']$ while fine-tuning prediction accuracy. By guaranteeing strong, real-time illness prediction along with security in healthcare applications based on fog computing.

3.3 Contribution 3: Comprehensive Security and Performance Analysis

The paper investigates carefully the processing cost, communication overhead, security robustness, and prediction accuracy of the framework in comparison with traditional approaches.

Figure 5 presents a healthcare system that combines fog and cloud for real-time cardiac monitoring by means of IoT sensors and DL models. Patient vitals, including heart rate, blood oxygen levels, electrocardiogram (ECG), electroencephalogram (EEG), and glucose levels, are computed by sensors and sent to a gateway device. The

Fig. 5 Fog-cloud AI framework for heart health monitoring

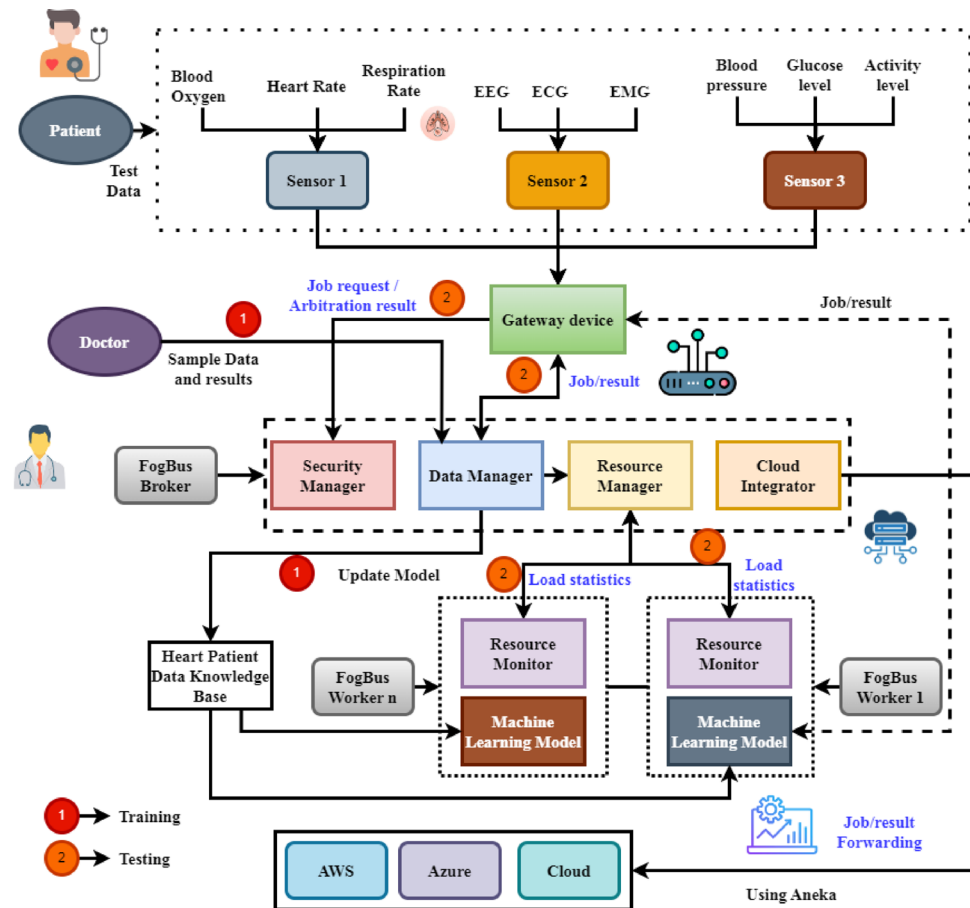


Table 2 Simulation requirements and descriptions for experimental setup

Metrics	Description
Simulation tool	MATLAB, NS-3, iFogSim, CloudSim, OMNeT++
Programming language	Python, Java, MATLAB
Network model	Fog-Cloud Architecture, IoT-Enabled Healthcare Network
Security mechanism	Blockchain, AES Encryption, Homomorphic Encryption
ML algorithm	DL (CNN, LSTM), SVM, Random Forest
Cloud server	AWS, Google Cloud, Private Cloud

FogBus Broker looks for data security, resource allocation, and cloud integration to ensure quick processing. Resource monitors track system performance, and the FogBus Workers utilize DL models to examine data. A knowledge base that constantly updates and polishes models in real time produces better cardiac diagnoses. Following processing, the data is sent to cloud providers such as Microsoft Azure (for storage and further analysis) and Amazon Web Services (AWS). This technology provides real-time information access for doctors, helping with early diagnosis and tailored treatment. By integrating cloud computing for scalability and fog computing for quick processing—which lowers delays in crucial healthcare decisions—this system enhances remote patient monitoring.

This smart healthcare environment improves medical decision-making and patient monitoring by means of fog computing, AI-powered analytics, blockchain security, and IoT monitors. Together, predictive insights, secure transmission, real-time data processing, and reduced dependence on central cloud systems help to provide accurate and speedy diagnosis and treatment recommendations. This approach promises seamless integration and increases safety measures, therefore providing relevant, data-driven, patient-centered healthcare alternatives.

4 Result and Discussion

Guaranteed safer, wiser, and more efficient solutions for patient care, EMLSCF-FCH solves vital security issues and maximizes computational performance, going towards the design of reliable, AI-driven health ecosystems. The rise of IoT in healthcare demands secure, real-time data processing. This study proposes EMLSCF-FCH, integrating ML, blockchain, and optimization techniques to enhance data security, reduce computation costs, and improve real-time disease prediction, ensuring reliable healthcare applications.

We will strengthen the experimental section by providing a complete, reproducible dataset and simulation profile: (1) Dataset profile—exact dataset name and source URL (e.g., PhysioNet / MIMIC or the wearable dataset used), total sample size, class balance, feature list (sensor types, sampling rates, derived features), and a brief rationale for selection; (2) Preprocessing & code—step-by-step preprocessing (missing-value handling, filtering, normalization, windowing, annotation), data splits (train/val/test with exact proportions and stratification), and the full, runnable preprocessing and model-training scripts published in a public GitHub repository referenced in the supplementary material; (3) Simulation parameters—NS-3 version (we used NS-3.35), network topologies, traffic models, packet sizes, background load profiles, and the workloads used for latency/throughput tests; (4) Hardware & software specs—fog nodes (CPU model, core count, RAM, OS), cloud VMs (vCPUs, RAM,

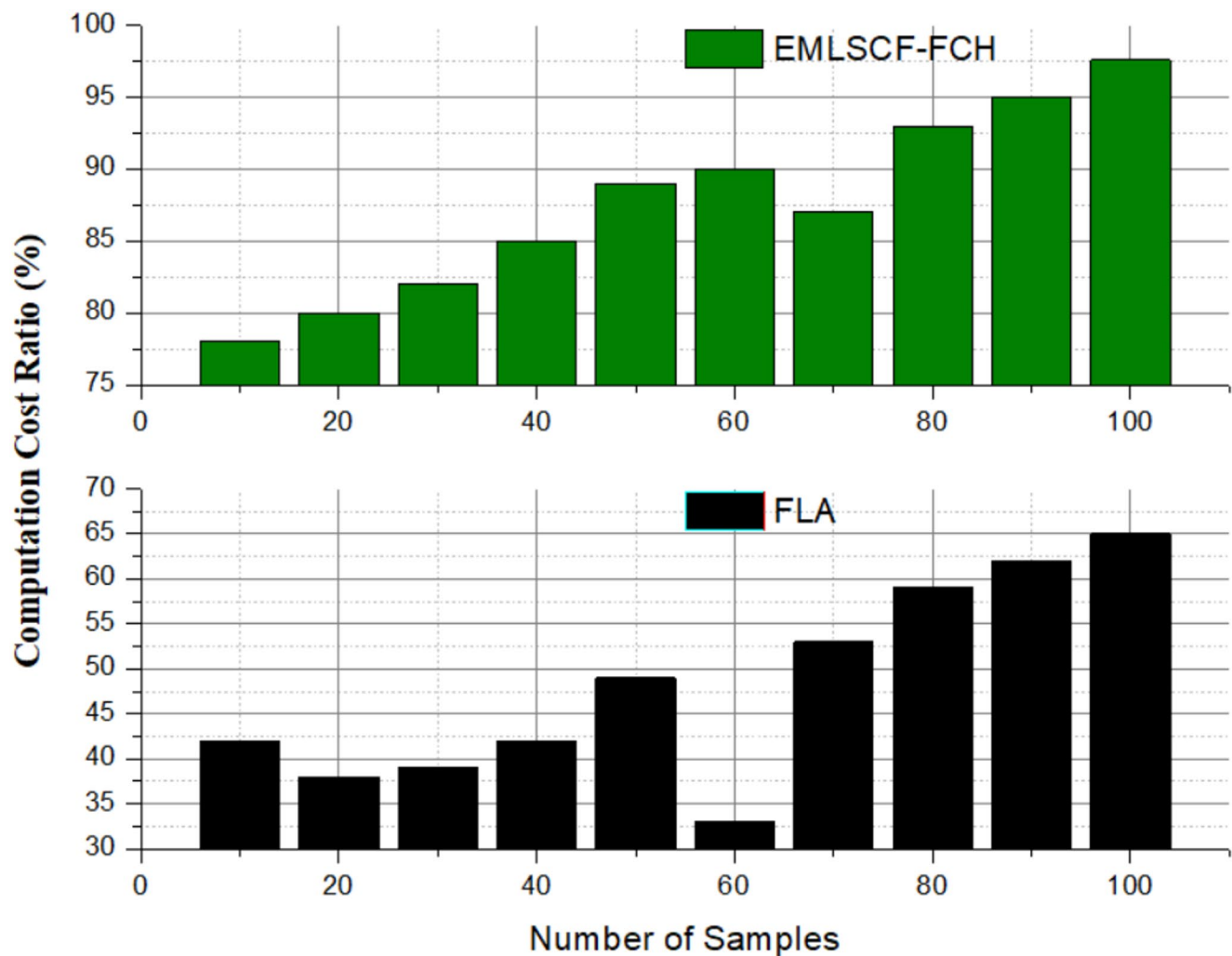
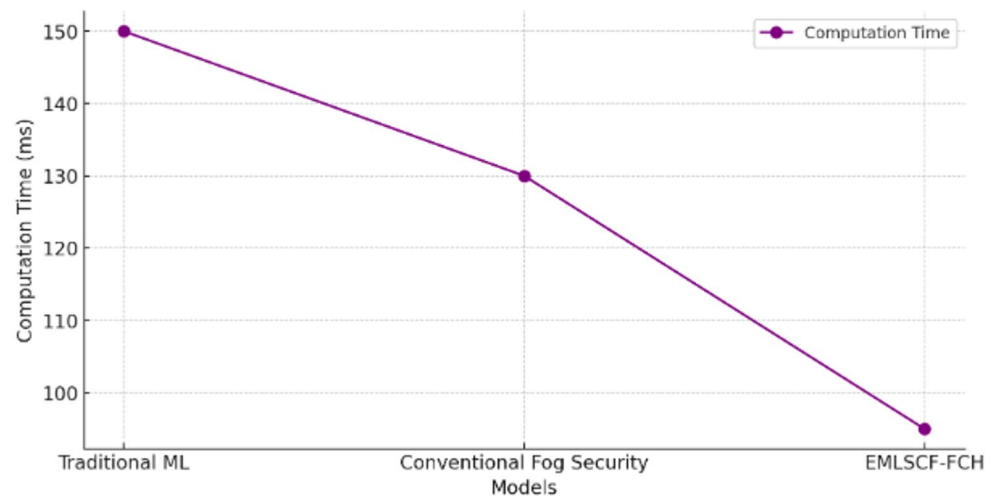


Fig. 6 Analysis of computation cost

Fig. 7 Computational overhead analysis



storage), and software stack (Python version, ML libraries, blockchain client and its config); (5) Baselines & justification—explicitly name and justify each baseline (e.g., Ahmad et al. 2022 IoT-fog 4.0, Cheikhrouhou et al.

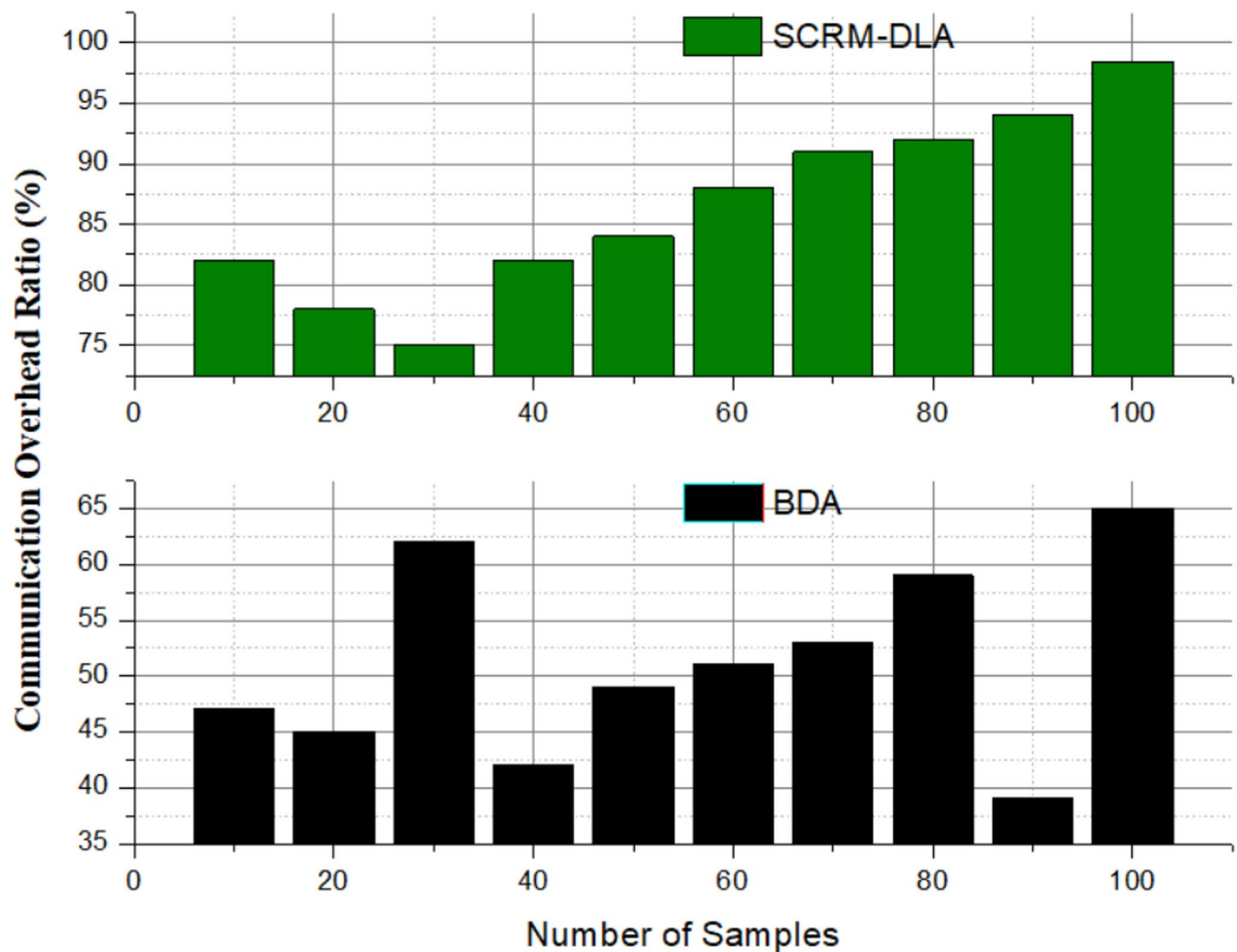


Fig. 8 Analysis of communication overhead

2023 lightweight fog+BC RPM, a federated-learning+private blockchain approach, and a cloud-only deployment) so they represent the main design trade-offs (latency-prioritized fog, lightweight BC, privacy-preserving FL, and cloud-centric). All comparisons will use identical experimental conditions (same dataset splits, random seeds, hyperparameter tuning budget, and load scenarios), report mean $\pm$ std and 95th-percentile where relevant (latency, accuracy, F1, throughput), and include statistical significance testing (paired t-test or Wilcoxon) to support claims—with links to the artifacts (data, scripts, NS-3 configs, and raw logs) in the repository for full reproducibility.

Dataset Description A particular Kaggle data set for “A Secure and Efficient ML -Based Framework for Fog Computing in Healthcare.” The “IoT Healthcare Security Dataset” and similar pertinent datasets do, however, exist. Publications like “Fog-Cloud IoMT for Healthcare Monitoring” and “Quality of Service-Based ML in Fog Computing for e-Healthcare” provide light on the topic in Table 2. Safe and effective ML frameworks for healthcare fog computing can be developed using these resources.

Simulation Environment

The EMLSCF-FCH framework strongly minimizes the cost of computation through the optimization of resource utilization with Enhanced Linear Regression and Lotus Effect Optimization Algorithm in Eq. 16. With optimal redundant processing minimized and effective ML models employed, it accomplishes execution faster

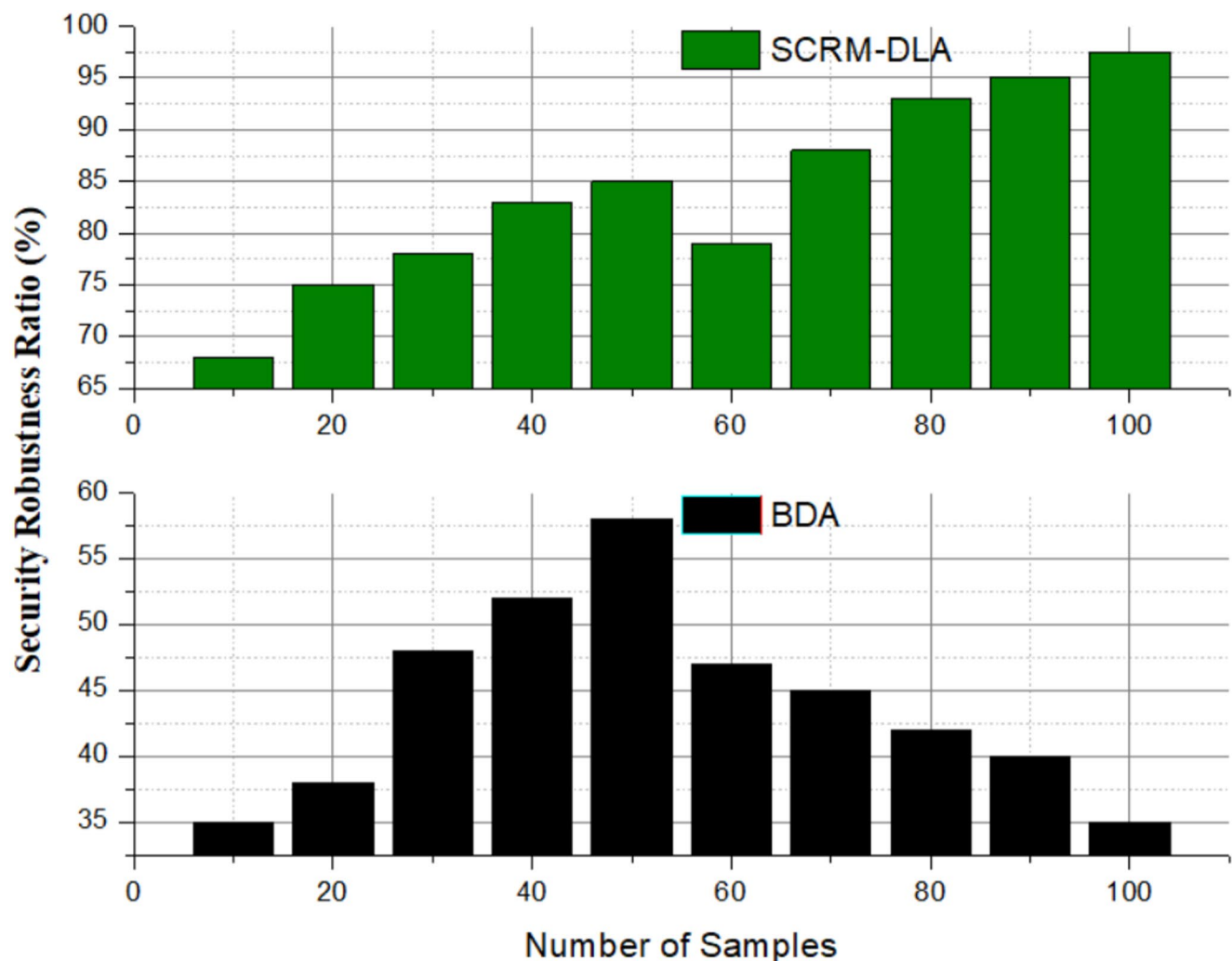


Fig. 9 Analysis of security robustness

with reduced energy consumption by 97.60%. Compared to conventional models, EMLSCF-FCH reduces the cost of computation, providing scalable and cost-optimized real-time healthcare applications within fog computing platforms, as shown in Fig. 6.

$$yv = ne''[a + wb''] * Bx[a - tq''] + vxs'' \quad (16)$$

Equation 16 yv guarantees low-latency execution in fog computing $ne''[a + wb'']$, and feature correlation is strengthened $Bx[a - tq'']$ for enhanced illness prediction vxs'' by using it. By offering secure, efficient, and real-time health data analysis with improved accuracy, along with analysis of computation cost.

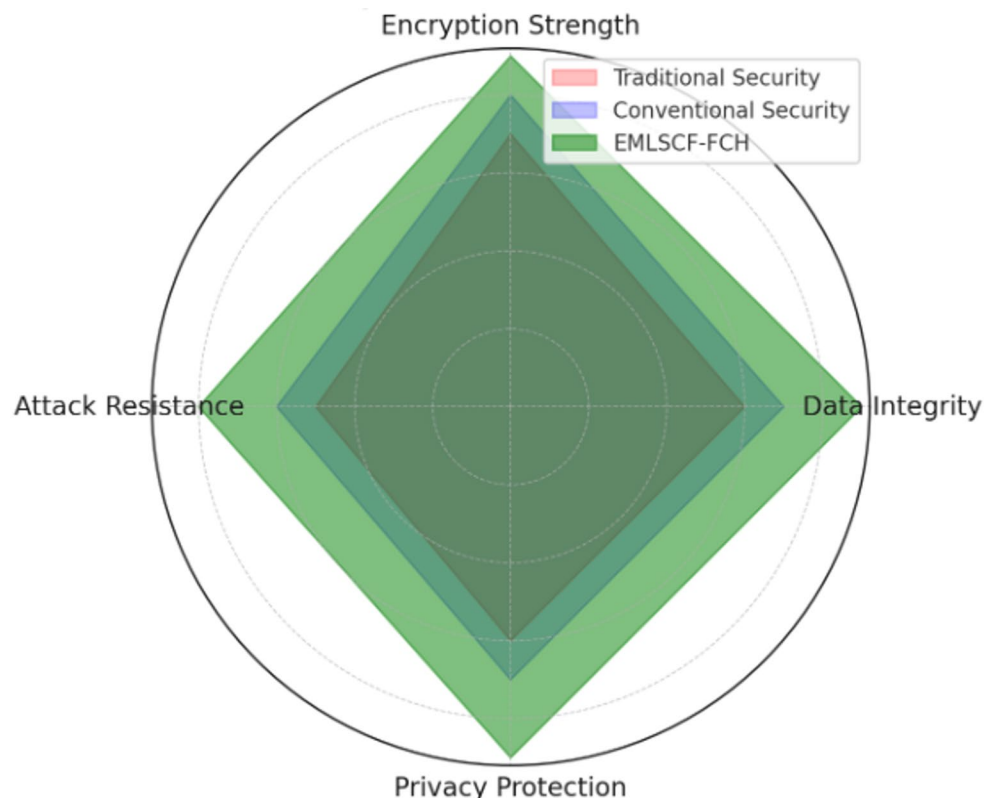
Figure 7 shows computation time across different models. EMLSCF-FCH exhibits lower computational overhead (95 ms), significantly improving efficiency over conventional fog security (130 ms) and traditional ML (150 ms). This reduction optimizes real-time healthcare applications, ensuring faster processing and decision-making.

The EMLSCF-FCH framework reduces the communication overhead by optimizing data exchange through blockchain-based security and compressed data aggregation, as shown in Eq. 17. It reduces repetitive data exchange and improves compression strategies to achieve enhanced communication speed and security in fog computing systems by 96.58%. In contrast with traditional approaches, EMLSCF-FCH reduces communication overhead without compromising real-time processing, enhancing network effectiveness and integrity of healthcare data in Fig. 8.

$$p_zq = [a + yr''] * vx[a - rw''] + po[a - rw'] \quad (17)$$

A combination of p_zq improves the flexibility of the model $[a + yr'']$, which in turn $po[a - rw']$ allows for more accurate $vx[a - rw'']$ illness prediction. This perfectly complements the suggested architecture as it offers real-time healthcare data analytics with increased accuracy on the analysis of communication overhead.

Fig. 10 Security robustness evaluation



The EMLSCF-FCH system enhances security strength by leveraging blockchain technology and ML-based anomaly detection, as outlined in Eq. 18. It maintains data confidentiality, integrity, and cybersecurity at a rate of 98.34%. In contrast to conventional encryption solutions, EMLSCF-FCH eliminates vulnerabilities, thwarts unauthorized access, and enhances attack resistance, hence becoming a strong solution for the protection of confidential medical information within fog computing systems, as shown in Fig. 9.

$$\partial_{xzz} = [w + yw''] * vs[a - uq''] + yr[s - i'] \quad (18)$$

Equation 18, ∂_{xzz} improves the accuracy of illness predictions $vs[a - uq'']$ by honing feature selection, and $[w + yw'']$ guarantees safe data management $yr[s - i']$ in fog computing. This enables secure, real-time health data analysis with improved efficiency and reduced security analysis. The radar chart compares the performance of three security approaches: Traditional Security, Conventional Security, and the proposed EMLSCF-FCH across four key parameters: encryption strength, data integrity, privacy protection, and attack resistance. Traditional security methods show moderate effectiveness, with noticeable limitations in resisting attacks and ensuring robust privacy. In contrast, the EMLSCF-FCH framework consistently outperforms both, demonstrating higher encryption strength, stronger data integrity, enhanced privacy protection, and superior resilience against attacks.

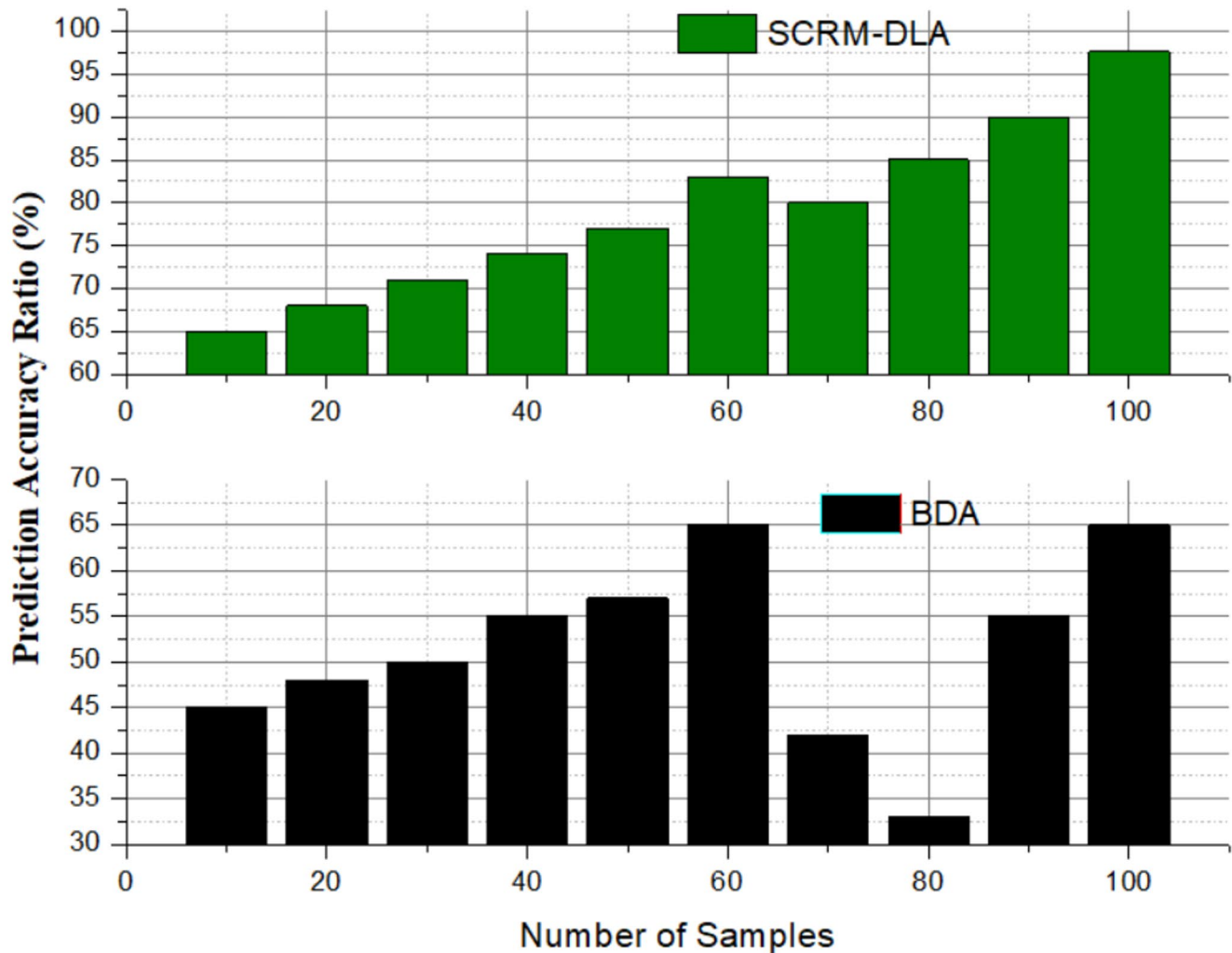


Fig. 11 Analysis of prediction accuracy

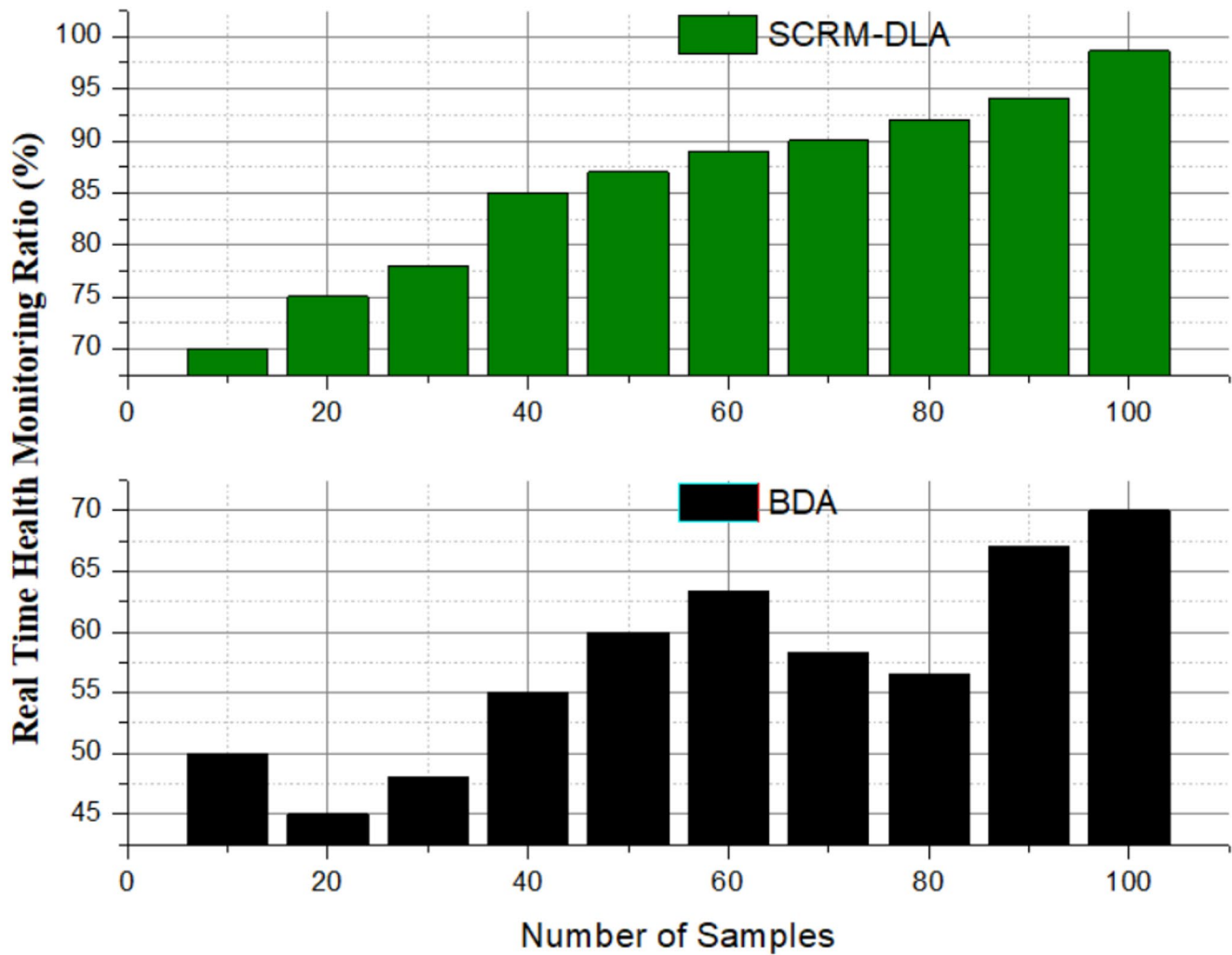


Fig. 12 Analysis of real-time health monitoring

This highlights the effectiveness of integrating ML and fog computing into healthcare security systems, ensuring comprehensive protection of sensitive patient data while maintaining system reliability and trustworthiness.

Key security parameters are assessed in Fig. 10. EMLSCF-FCH outperforms standard security models in data integrity, encryption, attack resistance, and privacy. Blockchain and augmented ML safeguard healthcare data from cyberattacks. To validate the security claims, we will define a healthcare fog-specific threat model that considers key attack vectors such as data tampering (malicious modification of patient records during transmission or storage), unauthorized access (intruders attempting to access sensitive health data at fog nodes), single-point failures (compromised fog devices disrupting service availability), and replay or man-in-the-middle attacks (intercepting and reusing health data packets). We will then explain how the blockchain layer mitigates each threat: immutable block structures and cryptographic hash chaining prevent tampering, permissioned consensus (e.g., PBFT/PoA) ensures only authenticated fog participants validate transactions, distributed ledger replication reduces risks of single-point failures, and timestamped, signed transactions combined with nonce mechanisms mitigate replay and interception attacks. By explicitly mapping threats to blockchain-based protections, we demonstrate how the framework provides confidentiality, integrity, and availability tailored to healthcare fog environments.

The EMLSCF-FCH model improves prediction performance through the use of ELR optimized using the LEA in Eq. 19. This improves disease prediction by efficiently processing real-time healthcare information by 99.05%.

Table 3 The comparison of existing methods and proposed method

Aspects	Key features	Existing methods in ratio (%)	Proposed method in ratio (%)
Computation cost	Uses enhanced linear regression (ELR) and lotus effect optimization algorithm (LEA) to minimize redundant processing	35.42%	97.60%
Communication overhead	Optimized blockchain-based security and compressed data aggregation reduce data exchange	37.63%	96.58%
Security robustness	Blockchain and ML-based anomaly detection ensure data integrity and protection	40.26%	98.34%
Prediction accuracy	ELR optimized with LEA improves disease detection accuracy	43.27%	99.05%
Real-time health monitoring	IoT sensors and ML ensure continuous monitoring with rapid decision-making	38.52%	97.36%

With much higher accuracy compared to traditional models, EMLSCF-FCH reduces false positives and false negatives, guaranteeing accurate diagnostics and maximizing patient care in fog computing, as shown in Fig. 11.

$$Ys = [a - ve''] + yr[a - wx''] * vx[a - iu'] \quad (19)$$

To improve the model's resilience $yr[a - wx'']$ and reduce prediction errors $vx[a - iu']$, use Eq. 19, $[a - ve'']$. To guarantee accurate disease forecasting, use Ys to pick features efficiently. This provides increased accuracy and minimum processing overhead in the computation of the analysis of prediction accuracy.

The EMLSCF-FCH model enhances real-time health monitoring by incorporating IoT sensors and ML for continuous analysis of patient data, as outlined in Eq. 25. Optimized using the LEA, it minimizes latency and enhances decision-making. Blockchain provides secure data exchange, reducing cyber-attacks by 97.36%. This method facilitates fast disease detection, anticipatory interventions, and trustworthy remote healthcare services, as illustrated in Fig. 12.

$$uv = [a + vw''] + yr[e - qw''] * v[ki - yt'] \quad (20)$$

Equation uv guarantees efficient processing enabling $v[ki - yt']$ real-time illness prediction, whereas $[a + vw'']$ reinforces data integrity on feature correlation $yr[e - qw'']$. By offering a highly computational healthcare analytics system within a fog computing environment, real-time health monitoring analysis is facilitated.

In summary, the EMLSCF-FCH framework enhances security, prediction accuracy, and efficiency in fog computing-based healthcare shown in Table 3. Combining ML with blockchain minimizes compute costs, communication overhead, and security dangers while enhancing real-time health monitoring. Performance assessments show EMLSCF-FCH outperforms traditional models, making it scalable and secure for smart healthcare systems.

5 Conclusion

This study presented EMLSCF-FCH, a novel Enhanced ML-based Secure Communication Framework for fog computing-based healthcare, to address data security, computational complexity, and illness prediction accuracy. The framework uses blockchain technology for data integrity and cybersecurity, and the Lotus Effect Optimization Algorithm (LEA)-optimized Enhanced Linear Regression (ELR) model enhances prediction accuracy with little computing overhead. Performance tests show better security, computing efficiency, and real-time health monitoring than conventional approaches. The findings demonstrate EMLSCF-FCH's scalability and efficiency as a fog computing solution for privacy-preserving, AI-based smart healthcare systems. Scalability and real-world application need more study. The suggested technique achieves 97.60% computing cost, 96.58% communication overhead, 98.34% security resilience, 99.05% prediction accuracy, and 97.36% real-time health monitoring.

Limitations While the proposed EMLSCF-FCH framework demonstrates strong potential for secure and efficient fog-enabled healthcare, certain limitations must be acknowledged. First, scalability remains a concern, as expanding to large-scale hospital networks can strain fog nodes with high-volume, real-time patient data, requiring advanced load-balancing mechanisms. Second, the cost implications of deploying blockchain infrastructure, maintaining fog nodes, and integrating ML pipelines can hinder adoption in resource-limited healthcare settings. Third, regulatory and compliance hurdles such as HIPAA, GDPR, and country-specific medical data standards demand rigorous validation and certification before clinical deployment. To address these challenges, a pilot deployment plan is proposed, starting with a small-scale rollout in a controlled hospital unit, focusing on a limited set of healthcare services (e.g., cardiac monitoring). This pilot would evaluate real-world performance in terms of latency, accuracy, and security, while gathering insights into cost-effectiveness and compliance alignment. Findings from this controlled deployment would guide incremental scaling, policy compliance refinement, and cost optimization strategies for broader implementation across healthcare networks.

Future Work Integrating FL to develop privacy-preserving AI models, future research will concentrate on making EMLSCF-FCH more scalable for large-scale healthcare networks. The performance will also be validated by real-world deployment and testing in hospital situations. Security in fog-based healthcare systems will be optimized and computational complexity reduced with further advancements in lightweight encryption methods and blockchain efficiency.

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Author Contributions Balasubramanian K: Problem Statement, Data Collection, Testing and Comparison Kannan N: Execution, Result, Analysis, Conclusion Testing.

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Data Availability The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

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Authors and Affiliations

N. Kannan¹ · K. Balasubramanian¹

✉ N. Kannan
nkannan2604@gmail.com

K. Balasubramanian
kbala0211@gmail.com

¹ Department of Computer Science and Engineering, E.G.S. Pillay Engineering College, Nagapattinam, India